Recoil and explosions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum conservation for an isolated system.

SUCCESS CRITERIA - I can...

1. Choose a system and check external impulse.
2. Add momentum with signs.
3. Explain equal-and-opposite momentum changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 kg cart travels left at 5 m/s. Find momentum and draw its direction.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{total,\,i} = p_{total,\,f}$.
Negligible net external impulse. Internal forces can change each part's momentum but not the system total.

Worked reasoning: Two carts initially at rest push apart with negligible external impulse. A 1 kg cart leaves at +4 m/s; a 2 kg cart must have momentum -4 kg·m/s, hence velocity -2 m/s.

Your turn: Draw before-and-after momentum arrows for the two-cart system.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two carts push apart from rest on an ideal track. Draw their momentum arrows. How must the two momenta compare?



A moves left B moves right

Marzano check - circle: ① ② ③

04 Apply the physics

A 1 kg cart moves left at 4 m/s after release. A 2 kg cart moves right. To keep total momentum zero, find the second cart's speed. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why can the two parts gain kinetic energy during a release while total momentum remains zero?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two balls launch horizontally from the same height at 4 and 24 m/s. Ignore air resistance. Compare fall times and horizontal ranges.
Label separate horizontal and vertical motion.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg cart leaves with momentum 6 kg·m/s left. What must the other cart’s momentum be if total was zero? Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Keep a 1 kg cart's departure velocity +6 m/s. Choose a second cart mass that recoils at -2 m/s. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3